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# CLINICAL COMMENTARY ON MIDFOOT AND FOREFOOT INVOLVEMENT IN LATERAL ANKLE SPRAINS AND CHRONIC ANKLE INSTABILITY. PART 2: CLINICAL CONSIDERATIONS

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## ABSTRACT

Lateral ankle sprains (LAS) and chronic ankle instability (CAI) are common musculoskeletal injuries that are a result of inversion injury during sport. The midfoot and forefoot is frequently injured during a LAS, is often overlooked during clinical examination, and maybe contributory to the development of CAI. The purpose of part two of this clinical commentary and current concept review is to increase clinician's awareness of the contribution of midfoot and forefoot impairment to functional limitation and disability of individuals who experience LAS and CAI and to facilitate future research in this area. The importance of multisegmented foot and ankle assessment from a clinical and research perspective is stressed. Select physical assessment and manual therapeutic techniques are presented to assist the clinician in examination and treatment of the ankle-foot complex in patients with LAS and CAI.

**Keywords:** Gait, intrinsic foot muscles, joint mobilization, physical examination, rehabilitation

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**Conflicts of Interest:** None

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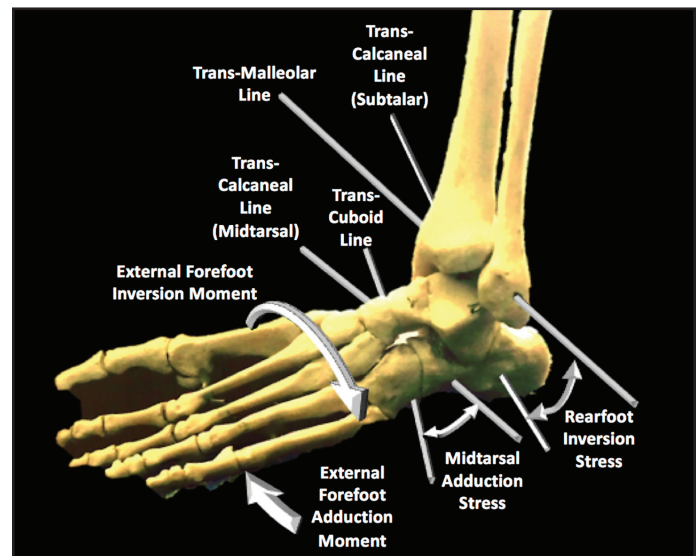
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## BACKGROUND AND PURPOSE

In the first part of this clinical commentary and current concepts review, foot and ankle anatomy, the roles of the intrinsic and extrinsic foot and ankle musculature from a multisegmented foot perspective, and the biomechanics of the ankle-foot complex during function were examined.<sup>1</sup> In part two of this this commentary, the contribution of midfoot and forefoot impairment in lateral ankle sprains and chronic ankle instability will be discussed in order to increase clinician's awareness and to facilitate future research in this area. The importance of multisegmented foot and ankle assessment will also be discussed from a clinical and research perspective.

Lateral ankle sprains (LAS) are common musculoskeletal injuries that affect more than two million individuals annually in the United States.<sup>2</sup> Only 11% of LAS patients perform supervised physical therapy following their injury.<sup>3</sup> Improper management of LAS may manifest into the residual impairment seen in the 40% of LAS patients that develop chronic ankle instability (CAI).<sup>4</sup> CAI is a chronic condition that involves impaired neuromuscular control, residual instability, and chronic pain that collectively result in self-reported disability after LAS.<sup>5-8</sup> Kinematic analyses of acute LAS's sustained during sport demonstrate rotational velocities up to 2124°/second which leads to extremes of range of motion, including up to 52° of plantarflexion, 126° of inversion, and 99° of adduction.<sup>9-13</sup> Simulated ankle sprains have demonstrated external moments in excess of 23 Nm for inversion and 11 Nm for adduction in simulated Grade I sprains.<sup>14</sup> LAS commonly involves damage to the anterior talofibular and calcaneofibular ligaments, which can be strained to approximately 20% and 16% of their resting length, respectively.<sup>14,15</sup>

Søndergaard<sup>16</sup> demonstrated that both the midfoot and forefoot are frequently injured during inversion ankle sprains and this phenomenon may be underappreciated by many clinicians. A number of midfoot injuries share similar mechanics to those incurred during a LAS.<sup>17-24</sup> Figure 1 depicts the external adduction and inversion moments that create lateral midfoot adduction stress and rearfoot inversion stress incurred during an inversion injury. The occult presentation of mild to moderate midfoot injury is likely attributed to the synchronicity of lateral



**Figure 1.** Lateral midfoot stress due to external adduction and inversion moments during an inversion injury.

ankle and midfoot injury. Inversion injuries frequently cause damage to the soft tissue structures of both the ankle and midfoot, while pain is often localized to the talocrural or subtalar articulations.<sup>16</sup> Nevertheless, if a patient reports inverting or 'rolling' their ankle, a thorough assessment of the lateral ankle joint and foot should simultaneously be performed. A recent clinical practice guideline published by the Orthopaedic Section of the American Physical Therapy Association recommends assessing patients who sustain LAS for painful foot conditions that may be indicative of fracture, cuboid involvement, or midfoot disruption.<sup>25</sup>

Despite improved understanding of the pathomechanics and pathophysiology of LAS and CAI, there is no evidence that the rate of recurrent LAS or CAI is declining. There is a need for further examination of other potential contributors to the etiology of recurrent LAS and CAI. Consideration of midfoot and forefoot involvement after LAS may be of clinical importance and the purpose of Part 2 of this clinical commentary and current concepts review paper is to increase clinicians' awareness of the contribution of midfoot and forefoot impairment to activity limitation and participation restrictions of individuals who experience LAS and CAI and to facilitate future research in this area. To accomplish this, the importance of multisegmented foot and ankle assessment from a clinical and research perspective will be reviewed.

## INJURIES INVOLVING THE MIDFOOT AND FOREFOOT

### Midfoot Injury

Midfoot injuries may include fractures, dislocations, subluxations, ligamentous sprains, or a composite of one or more of these injuries and are named by mechanism vector of the injurious force.<sup>26,27</sup> Examination of the foot is indicated when there is an apparent osseous or ligamentous injury in the foot. Prudence may dictate that the foot is examined in conjunction with the ankle following inversion injury, even when the patient does not report symptoms. The mechanism of midfoot injuries are frequently a consequence of ankle and foot supination that result in deleterious dorsal translation/axial compression in the medial column and plantar translatory/tensile distractive forces in the lateral column.<sup>28</sup> These injurious forces may culminate in ligamentous tears and osseous avulsions at the attachments of the calcaneocuboid, talonavicular or bifurcate ligaments.<sup>16</sup>

### Midfoot Injury in Lateral Ankle Sprains

In a prospective study of 711 patients who sustained an inversion sprain and were diagnosed in an urgent-care clinic, isolated midfoot sprains of either the bifurcate/dorsal calcaneocuboid ligament, talonavicular ligament, or both were found in 172 (26%) of the cases.<sup>16</sup> Additionally, midtarsal joint capsule involvement was found in 237 (33%) individuals who sustained LAS.<sup>16</sup> In another study investigating midfoot involvement in patients with history of LAS, damage to the bifurcate ligament was found in 40.5% of all cases.<sup>29</sup> Of these patients, 23% of the patients who had a diagnosis of "lateral ankle sprain," had isolated bifurcate ligament injury and an intact lateral ankle.<sup>29</sup> These findings illustrate that midtarsal joint injury is quite common, may mimic or contribute to lateral ankle signs and symptoms, and that the foot should be thoroughly examined following inversion ankle-foot injury. Because midtarsal joint injury may be misdiagnosed as a LAS, delay of care or improper clinical management may contribute to persistent activity limitation and participation restriction in these patients.

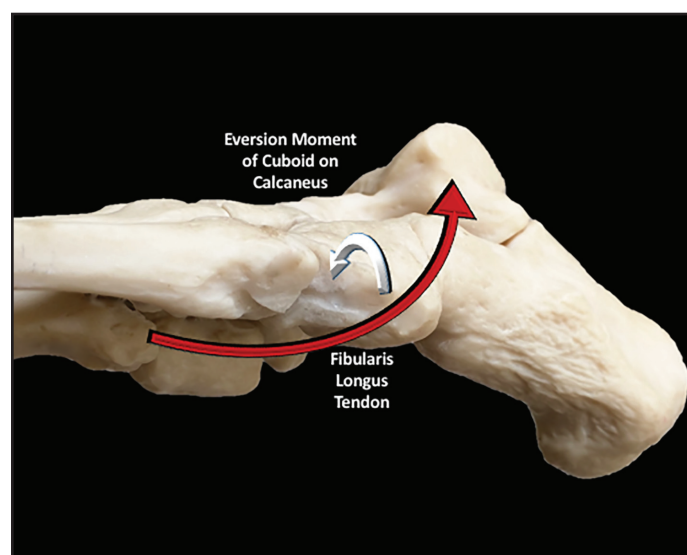
### Cuboid Syndrome

Cuboid syndrome, a lateral midfoot injury as a result of minor disruption of the calcaneal-cuboid congruency, has been described as being caused by abnormal inversion forces acting on the rearfoot when the fore-

foot is loaded during weight bearing.<sup>17</sup> Newell and Woodle<sup>30</sup> and Marshall<sup>31</sup> have described cuboid syndrome as a partial dislocation of the cuboid with subsequent impairment in motion. Similar mechanics have been described in multiple case studies of cuboid dislocation, a related and clinically more severe disruption of the calcaneocuboid articulation.<sup>18–24,32</sup> Analyses of case history and post-injury imaging have supported that when the forefoot is supinated, insult incurred from a dorsolateral external moment directed to the lateral aspect of the midfoot creates a plantar-medial displacement of the cuboid on the calcaneus.<sup>18–24,32</sup> It has been theorized that cuboid syndrome results from a calcaneocuboid subluxation created by forceful fibularis longus contractions during inversion injury.<sup>33</sup> The cuboid is normally everted and compressed during contraction of the fibularis longus as it courses around the fibularis sulcus.<sup>34</sup> During the combination of rearfoot inversion during forefoot loading, a medial and dorsal force vector created by the fibularis longus exerted on a medially rotated cuboid causes an infero-medial subluxation<sup>33</sup> (Figure 2). The subsequent discomfort associated with cuboid syndrome is attributed to the malposition of the cuboid and subsequent irritation of the joint capsules, ligaments, and the fibularis longus tendon.<sup>17</sup>

### Chronic Ankle Instability and the Multisegmented Foot

Many individuals who sustain LAS will subsequently develop persistent pathological gait kinematics<sup>35–39</sup>



**Figure 2.** Cuboid eversion with fibularis longus contraction.

and altered motor strategies<sup>37,40-42</sup> associated with CAI. CAI occurs in individuals that have had at least one significant ankle sprain, have repeat episodes of giving way, feelings of instability, or recurrent ankle sprains, and self-reported disability as a result of the ankle injury.<sup>5</sup> Groups of patients with CAI have been observed to walk with a wider bases of support,<sup>43</sup> decreased stride to stride variability in shank-rear-foot coupling,<sup>38</sup> increased shank external rotation excursion,<sup>39</sup> a more plantarflexed<sup>35</sup> and supinated foot,<sup>37,39</sup> and a more lateral center of plantar pressure progression<sup>41,42,44-46</sup> when compared to healthy controls. They have greater electromyographic activity for longer period of time in the gluteus medius and medial gastrocnemius pre initial contact (IC),<sup>42</sup> fibularis longus immediately pre<sup>36,37,41,42</sup> and post<sup>36,37,41</sup> IC, and gluteus medius from 50% of stance phase to 25% of swing phase.<sup>42</sup> Evidence is conflicting regarding the electromyographic activity in the tibialis anterior during the stance phase of gait, with both increased<sup>41</sup> and decreased<sup>42</sup> activity reported. Impairment in the midfoot<sup>47,48</sup> and medial forefoot kinematics<sup>49</sup> have been suggested to be contributory in CAI. Interestingly in a study of 711 patients who sustained inversion injury isolated to either the lateral ankle (65% of sample) or the midtarsal joint (23% of sample), pain, swelling, perception of giving way, and subsequent inversion injury persisted at the same frequency at 6-12 months regardless of the site of injury.<sup>16</sup>

In order to make the case of suspected midfoot involvement in CAI, there are some recent studies in healthy subjects that may provide some contrast and relevance. A study of healthy individuals who were classified as having a large inversion forefoot angle at IC ( $5.9 \pm 1.6^\circ$ ) were found to have a greater forefoot pronation excursion and remain everted for longer periods during stance when compared to a group who had a moderate forefoot angle ( $2.6 \pm 1.1^\circ$ ) at IC.<sup>50</sup> Similarly, the findings of a kinematic study of the rearfoot coupling mechanism were that healthy midtarsal joints uncoupled from the rearfoot post IC and remained unlocked through terminal stance.<sup>51</sup> This finding challenges the notion that the midtarsal joint locks the rear and midfoot at terminal stance in order to provide a rigid lever required for efficient gait.

Groups of patients with CAI have been found to have up to  $7^\circ$  more inversion in the rearfoot prior to and

following IC when compared to healthy controls, which has been suggested to be a contributing factor to this population's increase risk for reinjury.<sup>36,37</sup> Morphologically a group of patients with CAI who were scheduled for lateral ankle reconstruction were found on radiograph to have significantly higher mean talometatarsal and talocalcaneal angles, and lower mean calcaneal angles and tarsal indices when compared to healthy controls, indicating higher medial longitudinal arches and cavovarus.<sup>52</sup> It has been suggested that cavovarus in patients with CAI is a major contributing factor in the progression to ankle osteoarthritis and corrective calcaneal osteotomy should be considered in conjunction with ligamentous reconstruction to normalize forces about the ankle-foot complex.<sup>53</sup>

Changes in plantar pressure during walking have been found in patients with CAI when compared to healthy controls.<sup>41,44-46,54</sup> Nyska and colleagues<sup>45</sup> found patients with CAI spend more time in the rear and midfoot during stance with a delay in transition to the central and lateral forefoot and toes, increased pressure in the midfoot and lateral forefoot, and decreased pressure in the heel and toes. Schmidt and colleagues<sup>46</sup> also found a delay in time to peak pressure of the medial and lateral rearfoot and the medial midfoot during early stance phase in patients with CAI. Patients with CAI have greater plantar pressure under the midfoot and lateral forefoot and decreased pressure in the heel and toes compared to healthy controls.<sup>45,46</sup> Nawata and colleagues<sup>44</sup> found that patients with CAI ambulated with a laterally deviated center of progression and an adducted/supinated foot during midstance. Hopkins and colleagues<sup>41</sup> observed similar findings with subjects with CAI walking with increased lateral center of pressure progressions between 20% to 90% of stance when compared to healthy matched controls. Koldenhoven and colleagues<sup>42</sup> found that patients with CAI have a more lateral center of pressure progression throughout the stance phase and have increased plantar pressure in the lateral forefoot for longer periods of time. Individuals with CAI also run with a lateral plantar pressure distribution during foot strike and the plantar progression starts more laterally during initial loading when compared to controls.<sup>54</sup>

The kinematic and kinetic findings found in patients with CAI may result from the impaired ability to

uncouple the midfoot from the rearfoot due to mechanical or neurophysiologic constraints. Impaired joint mobility in any of the foot segments may impair the ability of the foot to decouple during lower velocity ambulation. A neurophysiologically constrained midfoot combined with a supinated rearfoot could plausibly contribute to the lateral shift in plantar center of pressure progression during the stance phase of gait. Joint mobility assessment and manipulation has been recommended in clinical cases of idiopathic cavovarus, especially when associated with gait abnormalities and clinical entities such as LAS and ankle instability.<sup>55</sup>

Hypomobility of the first ray may contribute to the lateral shift in plantar pressure seen in this patient population.<sup>56</sup> It is plausible that joint hypomobility could also affect the muscles acting on the first ray and may explain the findings of a recent study, where patients with CAI were found to have atrophy of the flexor hallucis brevis and flexor hallucis obliquus and hypertrophy of the flexor hallucis longus.

Neuromuscular adaptations in the foot such as co-contraction of the extrinsic and intrinsic antagonistic pairs may also be implicit in CAI. Increased muscle stiffness is thought to be beneficial in joint stability, especially when mechanical stability is impaired and the muscles play a larger role in mitigating destabilizing forces.<sup>57</sup> If there is mechanical disruption of the transverse tarsal, tarsometatarsal, or intertarsal ligamentous structures, it is plausible that stabilizing co-contraction in the foot may create a situation where the rearfoot, midfoot, and forefoot remain coupled throughout stance, creating a constrained system.

Impaired coupling in the foot may also occur in the CAI population due to neuromuscular dysfunction of the extrinsic and intrinsic musculature. In electrophysiological studies of 66 patients who sustained LAS, Nitz and colleagues<sup>58</sup> found decreased nerve conduction velocities in the peroneal (17% of patients with a Grade II LAS, 86% of patients with a Grade III LAS) and tibial (10% of patients with a Grade II LAS, 83% of patients with a Grade III LAS) nerves, as well as electromyographic evidence of denervation. Jazayeri and colleagues<sup>59</sup> also found increased peroneal and tibial nerve latencies during nerve conduction studies of football (soccer)

players who sustained LAS. Impaired fibularis longus or intrinsic foot muscle function secondary to neuropraxia or traction axonotmesis/neurotmesis may be deleterious to intersegment coupling, foot shaping, intersegmental stability, force attenuation, and afferent feedback from the articular soft tissue and plantar cutaneous sensation. In the only study known to investigate individuals with CAI utilizing a multisegmented foot model during walking, the first ray was found to have a mean 9.4° more inversion from 87% to 98% of stance phase when compared to healthy controls.<sup>49</sup> Similar findings were observed in LAS copers, operationally defined as subjects who had sustained LAS in the previous two years but were not experiencing ankle instability, had a mean 7.4° difference from 10% to 83% of stance phase.<sup>49</sup>

The fibularis longus, besides being an extrinsic everter of the foot, is a plantarflexor and everter of the hallux, and stabilizes the medial column, medial longitudinal and transverse arches<sup>60</sup> and the calcaneocuboid joint.<sup>34</sup> Impaired peroneal function has been offered as a possible explanation for the supinated position of the hallux in patients with CAI.<sup>49</sup> Patients with CAI have been found to have decreased concentric and eccentric strength,<sup>61</sup> diminished mean activation time,<sup>62</sup> and increased latency and electromechanical delay<sup>63</sup> in the fibularis longus in the injured limb when compared to healthy controls. Due to the proximity of the fibularis longus to the cuboid, minor disruption in cuboid congruency or subluxation is thought to contribute to peroneal irritability<sup>64</sup> and may contribute to impaired function of this muscle. The cuboid functions as a pulley for the fibularis longus tendon and provides a more advantageous vector of pull to support the transverse arch, medial longitudinal arch, and the first ray.<sup>60</sup> More substantial disruption in stability or position of the calcaneus may have the potential to disrupt this pulley mechanism by altering tendon slack length or the vector of force. Patients with CAI have been found to walk at lower velocities<sup>43</sup> and with an adducted foot.<sup>45</sup> It is plausible that impaired ability to lock the midfoot due to ligamentous instability or neuromuscular impairment in the fibularis longus may force patients with CAI to employ a gait strategy where pushoff occurs about the oblique metatarsal axis. This may also explain some of the plantar pressure findings found in the lateral forefoot in patients with

| <b>Table 1. Observational and Clinical Measures of Foot Morphology.</b>                                                                                                                                                                   |                                                                |                                                                                                                                                                                                                                  |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Segment</b>                                                                                                                                                                                                                            | <b>Assessment</b>                                              | <b>Summary</b>                                                                                                                                                                                                                   |
| Multi-segmented                                                                                                                                                                                                                           | Foot Posture Index – 6 item                                    | A composite measure of foot posture that is based on one palpation and five observations of foot morphology in standing.                                                                                                         |
| Longitudinal Arch (Midfoot)                                                                                                                                                                                                               | Change in Foot Length (NWB to FWB)                             | A measure of change of total foot length in both NWB and FWB conditions.                                                                                                                                                         |
|                                                                                                                                                                                                                                           | Δ Arch Height Index (Navicular height/Foot Length, NWB to FWB) | A measure of change of navicular height normalized to foot length in both NWB and FWB conditions.                                                                                                                                |
| Longitudinal and Transverse Arches (Midfoot)                                                                                                                                                                                              | Δ Dorsal Arch Height (NWB to FWB)                              | A measure of change in vertical arch height between loading and unloading. The dorsal aspect foot is measured at 50% of total foot length in both NWB and FWB conditions.                                                        |
|                                                                                                                                                                                                                                           | Δ Midfoot Width (NWB to FWB)                                   | A measure of change of midfoot width between loading and unloading. The width of the foot is measured at 50% of total foot length in both NWB and FWB conditions.                                                                |
|                                                                                                                                                                                                                                           |                                                                | $\text{Foot Mobility Magnitude} = \sqrt{(\Delta \text{ height})^2 + (\Delta \text{ width})^2}$                                                                                                                                   |
|                                                                                                                                                                                                                                           | Foot Mobility Magnitude                                        | A composite measure derived from change of midfoot height and width. We posit that this may be an acceptable surrogate measure for the assessment of intrinsic/extrinsic muscular control of the foot across loading conditions. |
| Subtalar (rearfoot)                                                                                                                                                                                                                       | "Too Many Toes" sign                                           | An observational assessment of the pronated foot during standing, when the lateral forefoot/lesser toes can be observed when viewing the foot from an posterior-anterior perspective                                             |
|                                                                                                                                                                                                                                           | "Peak-a-Boo" sign                                              | An observational assessment of the supinated rearfoot during standing, when the medial calcaneus can be observed when viewing the foot from an anterior-posterior perspective                                                    |
| NWB= non-weight bearing, FWB= full weight bearing                                                                                                                                                                                         |                                                                |                                                                                                                                                                                                                                  |
| The Foot Posture Index is based on the work of Redmond AC, Crosbie J, Ouvrier RA. Development and validation of a novel rating system for scoring standing foot posture: The Foot Posture Index. Clinical Biomechanics. 2006;21(1):89-98. |                                                                |                                                                                                                                                                                                                                  |

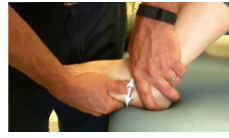
CAI. This gait strategy may also be utilized to maximize balance in the presence of other neurophysiologic impairment.

## CLINICAL IMPLICATIONS AND FUTURE DIRECTION

The midfoot plays an essential role in force transmission during gait, is commonly injured during inversion sprains, and is likely to contribute to the morbidity associated with LAS and CAI. Clinically, it is important to consider the midfoot and forefoot during examination and treatment of these patients. It has been previously suggested that the diagnostic scope should be widened to include the midfoot when assessing and treating common ankle sprains.<sup>16</sup> Based on the evidence presented in this paper, it is recommended that patients may benefit from examination of the midfoot and forefoot post inversion injury, even when the patient does not report pain symptoms in the region. If treating providers fail to assess the midfoot and forefoot following LAS, it is likely that important contributory impairment will be missed.

The authors recommend that clinicians take a holistic approach when examining and performing treatment in those who sustain LAS. A detailed clinical history that captures type and duration of symptoms, recurrence, mechanism of injury, timing and location of pain complaints, and current functional limitations will help guide the physical examination. Inquiry to factors, that when implemented have been shown to hypertrophy the intrinsic foot muscles and beneficially modify foot shape, such as minimalist footwear<sup>65,66</sup> time spent barefoot,<sup>67–69</sup> and the type of surface physical activity occurs (outdoors > indoors)<sup>67</sup> may provide the clinician insight regarding intrinsic foot strength. Observation of foot morphology, in both unloaded and loaded conditions, can provide information on the patient's ability to shape and stabilize the foot. Measurements of navicular height, dorsal arch height, foot length, and foot width in both loading conditions are expedient and clinically meaningful methods of assessing control of the longitudinal and transverse arches. Table 1 presents some suggested observational and clinical measures of foot morphology.

**Table 2.** Joint Mobility Assessment of the Ankle-Foot Complex. Joints are graded as having normal mobility, hypermobility, or hypomobility

| Assessment                                            | Procedure                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Image                                                                                 |
|-------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| 1 <sup>st</sup> Metatarso-phalangeal Passive Mobility | The patient is positioned in hook lying with the toes cantilevered over the plinth edge. With the the distal 1 <sup>st</sup> metatarsal stabilized, the proximal phalanx is distracted and glided in a plantar vector parallel to the joint line until the end feel is noted. Assessment is repeated utilizing a dorsal glide. Annotate joint laxity using the passive mobility scale.                                                                                                                |    |
| Hallux Extension Excursion                            | The patient is positioned in hook lying with the feet flat on the plinth. With the 1 <sup>st</sup> metatarsal stabilized, the metatarsophalangeal joint is passively extended until the end feel is noted. Goniometric measurement and joint end feel is recorded.                                                                                                                                                                                                                                    |    |
| 1st Tarso-metatarsal Plantar glide                    | The patient is positioned supine with the distal shank cantilevered over the plinth edge. The distal 1 <sup>st</sup> metatarsal is stabilized with the assessor's thumb pads. The pads of the assessor's middle and index fingers are placed on the distal segment of the dorsal first tarsometatarsal joint. Force is applied using the overlapped fingers from the assessor's opposite hand in a plantar vector parallel to the joint line. Annotate joint laxity using the passive mobility scale. |    |
| 1st Tarso-metatarsal excursion                        | The patient is positioned supine. The most distal aspect of the 1 <sup>st</sup> cuneiform is stabilized while the 1 <sup>st</sup> metatarsal is passively dorsiflexed until the end feel is noted. Goniometric measurement and joint end feel is recorded. The measure is repeated for plantarflexion.                                                                                                                                                                                                |    |
| Forefoot and Midfoot Inversion/Eversion Mobility      | The patient is positioned supine with the distal shank cantilevered over the plinth edge. The stabilizing hand cradles the calcaneus to prevent rearfoot inversion/eversion. Utilizing a "C" grip, grasp the forefoot at the metatarsal heads. The forefoot is maximally inverted until terminal excursion is achieved. End feel and goniometric measurement is recorded. Repeat procedure for eversion.                                                                                              |   |
| Midfoot Adduction passive mobility                    | An assessment of calcaneocuboid and calcaneonavicular ligamentous and capsular laxity/provocation. The patient is positioned supine with the distal shank cantilevered over the plinth edge. The stabilizing hand cradles the calcaneus to prevent rearfoot movement. An adduction force is exerted to the lateral forefoot. Annotate joint laxity using the passive mobility scale.                                                                                                                  |  |
| Talonavicular passive mobility                        | The patient is positioned in hook lying with the feet flat on the plinth. With the dorsal rearfoot stabilized, the navicular is passively glided in a dorsal vector until the end feel is noted. Assessment is repeated utilizing a plantar glide. Annotate joint laxity using the passive mobility scale.                                                                                                                                                                                            |  |
| Calcaneo-cuboid passive mobility                      | The patient is positioned in hook lying with the lateral foot cantilevered over the edge of the plinth. With the dorsal rearfoot stabilized, the cuboid is passively glided in a dorsal vector until the end feel is noted. Assessment is repeated utilizing a plantar glide. Annotate joint laxity using the passive mobility scale.                                                                                                                                                                 |  |
| Subtalar eversion passive mobility                    | The patient is positioned in a side lying position on the side that is to be assessed and cantilevered off the edge of the plinth. The shank and talus is stabilized. With the calcaneus cradled, a lateral force is exerted through the medial calcaneus. Annotate joint laxity using the passive mobility scale.                                                                                                                                                                                    |  |

Palpatory examination of the joints, ligaments, and muscles of the foot is important post inversion injury to assist in determining midfoot or forefoot involvement. Joint range of motion and accessory motion assessment in each segment and joint of the

foot will often reveal intersegmental joint limitation and provide the clinician with a prime opportunity to render treatment such as joint mobilization or manipulation. Suggested manual therapeutic techniques for the ankle-foot complex are presented in

**Table 3. Joint Manipulation of the Midfoot and Medial Forefoot**

| Segment                | Intervention                                       | Indication                                              | Procedure                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Picture                                                                               |
|------------------------|----------------------------------------------------|---------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| Medial Forefoot/Hallux | Hallux (MTP) Dorsal Glide                          | Impaired dorsiflexion of the hallux                     | The patient is positioned supine in hook lying. The distal metatarsal is stabilized between the pad of the thumb and the proximal phalanx of the index finger (use of a piece of tubigrip of latex glove may assist in maintaining grip). The base of the proximal phalanx is gripped using a similar method. A grade III axial distractive force is exerted to achieve joint separation. A plantar force is exerted on the proximal segment while simultaneously exerting a dorsal counterforce on the distal segment to take up joint slack. Once at the end of the joint excursion, a high frequency, low amplitude oscillatory grade IV manipulation may be applied to the joint. |    |
|                        | 1 <sup>st</sup> TMT Plantar Glide                  | Impaired dorsiflexion of the 1 <sup>st</sup> metatarsal | The patient is positioned supine. The treating clinician will wrap the secondary hand around the lateral aspect of the foot perpendicular to the proximal 1st metatarsal. The pads of the middle and index finger will rest adjacent and just distal to 1st tarsometatarsal joint. The primary hand will wrap over the medial aspect of the foot and overlay the fingers of the secondary hand. A plantar direct force is exerted through the proximal metatarsal to take up joint slack and to introduce dorsiflexion. Once at the end of the joint excursion, a high frequency, low amplitude oscillatory grade IV manipulation may be applied to the joint.                        |    |
| Midfoot                | Dorsolateral Cuboid Glide with Forefoot Supination | Hypomobility or disrupted calcaneo-cuboid congruency    | The patient is positioned supine. The heel is cupped and the plantar aspect of the cuboid blocked by the treating clinician's stabilizing hand. The clinician grips the forefoot across the metatarsal heads and exerts a supination force, taking up the slack of the multiple segments from the midfoot to forefoot. At terminal excursion, overpressure is applied to supination while providing a dorsolateral counter force to the cuboid.                                                                                                                                                                                                                                       |   |
|                        | Calcaneo-cuboid Squeeze                            |                                                         | The patient is positioned prone with the knee flexed to 70°. The patient's foot is gripped with both hands and the pads of both thumbs contacting the plantar aspect of the cuboid. With constant dorsal pressure applied to the cuboid, the ankle is slowly stretched into terminal plantarflexion with the midfoot and toes maximally flexed.                                                                                                                                                                                                                                                                                                                                       |  |
|                        | Cuboid Whip                                        |                                                         | The patient is positioned prone with the knee flexed to 70-90° and the talocrural joint in neutral. The patient's foot is gripped with both hands and the pads of both thumbs contact the plantar aspect of the cuboid. With constant dorsal pressure applied to the cuboid, the knee is rapidly extended and the ankle rapidly plantarflexed.                                                                                                                                                                                                                                                                                                                                        |  |
|                        | Talonavicular Plantar and Dorsal Glides            | Talo-navicular/medial longitudinal arch hypomobility    | The patient is positioned supine in hook lying. The treating clinician will wrap the fingers of the stabilizing hand around the talus, distal to the malleoli. The mobilizing hand will grasp the navicular with the pad of the thumb in contact with the dorsal navicular. With the talus stabilized a plantar or dorsal force may be exerted to take up joint slack. Once at the end of the joint excursion, a high velocity, low amplitude force or a high frequency, low amplitude oscillatory grade IV manipulation may be applied to the joint.                                                                                                                                 |  |

MTP=metatarsal phalangeal, TMT= tarsometatarsal

Tables 3 and 4. In the cases of segmental instability, the plan of care can be modified to allow for protection, intervention such as taping/strapping, bracing, orthotic fitting, and foot core stabilization exercises, and/or referral to orthopedic surgery or podiatry for

surgical consideration. Table 2 presents some suggested joint mobility assessment techniques that can be used in the clinical examination.

Assessment of intrinsic muscle function can be difficult without the use of laboratory equipment or imag-

**Table 4. Joint Manipulation of the Rearfoot and Shank**

| Segment               | Intervention                                           | Indication                                                         | Procedure                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Picture                                                                               |
|-----------------------|--------------------------------------------------------|--------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| Subtalar              | Subtalar Lateral Glide                                 | Impaired rearfoot eversion                                         | The patient is positioned sidelying, on the side of the injured limb. The treating clinician will wrap the fingers of the stabilizing hand around the talus, distal to the malleoli. Using the thenar eminence of the manipulating hand as a contact to the medial calcaneus, the clinician will apply a downward force through the arm into the patient's medial calcaneus to take up any joint slack. Once at the end of the joint excursion, a high velocity, low amplitude force or a high frequency, low amplitude oscillatory grade IV manipulation is applied to the joint.                                                                                                    |    |
| Talo-crural           | Talar Distraction Manipulation                         | Impaired talocrural dorsiflexion or plantarflexion                 | With the patient positioned supine, the treating clinician places both hands with the fingers intertwined on the most posterior aspect of the dorsal foot, closest to the axis as possible. Using the thumbs, the clinician will position and maintain the foot in 10° of plantarflexion. The clinician will take up the joint slack by applying a caudal distractive force to the foot. Once at the end of the joint excursion, a high velocity, low amplitude force is exerted.                                                                                                                                                                                                     |    |
|                       | Posterior Talar Glide Manipulation                     | Impaired talocrural dorsiflexion                                   | The patient is positioned supine with the foot extended beyond the edge of the plinth. The treating clinician will use the soft tissue of the web space and the thenar eminence between the thumb and index finger of the manipulating hand to contact the anterior talus just distal of the malleoli. The clinician will position and maintain the foot in 10° of plantarflexion and a stabilizing force to the shank while applying an anterior-posterior force to the talus to take up the joint slack. Once at the end of the joint excursion, a high velocity, low amplitude force or a high frequency, low amplitude oscillatory grade IV manipulation is applied to the joint. |    |
| Tibio-fibular (shank) | Proximal Tibiofibular Joint Manipulation               | Impaired fibula translation during ankle dorsiflexion and eversion | The patient is positioned supine with the hip and knee flexed. The treating clinician wraps one hand around the lateral aspect of the proximal shank into the popliteal space, with palmar surface of the metatarsophalangeal joint of digits 1 & 2 contacting the patient's fibular head. The shank is externally rotated and the knee is flexed until joint slack is taken up at the proximal tibiofibular articulation. Once at the end of the joint excursion, a high velocity, low amplitude force is exerted at the distal end of the shank.                                                                                                                                    |  |
|                       | Distal Tibiofibular Joint Posterior Glide Manipulation |                                                                    | The patient is positioned supine with the heel cupped in the treating clinician's stabilizing hand. The clinician contacts the anterior lateral malleolus using the thenar eminence of the mobilizing hand and applies an anterior-posterior force to take any joint slack. A high frequency, low amplitude oscillatory grade IV mobilization is applied at the end of joint excursion.                                                                                                                                                                                                                                                                                               |  |

ing modalities that are either not feasible or accessible for regular clinical use. Equipment such as motion capture systems, electromyographic, or magnetic resonance imaging machines is expensive, take clinical space, or require time-consuming technical analysis. Clinically, there are strategies that practitioners may use to objectively collect surrogate measures of intrinsic muscle function. The intrinsic muscles have been found to have the ability to control deformation of the longitudinal arch.<sup>70</sup> Measurement of the navicular height, foot length, and width using a tape

measure or caliper in both unloaded position and in standing is a time expedient and inexpensive method. Toe flexor strength has been found to be associated with cross sectional area of both the extrinsic and intrinsic foot muscles, with larger size of the medial plantar intrinsic foot muscles (flexor hallucis brevis, flexor digitorum brevis, quadratus plantae, lumbricals and abductor hallucis) being a major predictor of toe flexor strength.<sup>71</sup> Manual muscle testing of the toe flexors is an easy and quick assessment technique that may yield clinically relevant information about

the intrinsic foot muscles. Testing the patient's ability to isolate great toe movements from the lateral forefoot (great toe abduction, great toe extension with flexion in toes 2-5, great toe flexion with extension in toes 2-5) and strength testing may yield pertinent information on intrinsic function.

Ultrasound imaging is a modality that has emerged in the literature for the assessment of intrinsic foot muscle size.<sup>72-76</sup> Many clinicians have access to ultrasound imaging units, which makes this imaging modality ideal for use in evaluation of patients with ankle-foot pathology. Clinicians may find ultrasound imaging useful as an outcome measure for tracking changes in resting muscle size to assess effectiveness of exercise intervention (hypertrophy) or atrophy following disuse or neuromuscular insult. There is also great potential for the use of this imaging modality for the assessment of neuromuscular function or as a bio-feedback instrument. Assessment of the intrinsic and extrinsic muscles of the ankle-foot during a state of contraction may provide a clinician great insight to neuromuscular function and motor control. Future research is needed to establish the measurement properties of ultrasonography for assessment of neuromuscular function of the intrinsic foot muscles.

Testing of extrinsic muscle function of the ankle-foot complex is a standard of care when treating patients with LAS or CAI. Commonly, assessment is comprised of manual muscle testing (MMT) or hand held dynamometry of the open kinetic chain motions of ankle dorsiflexion, plantarflexion, inversion, and eversion. While MMT is a convenient assessment technique that may reveal information about single segment, open kinetic chain function of the extrinsic muscles, they may not translate well to how these muscles function in relation to the multisegmented foot. It has been recommended that strength testing and training should be specific with consideration given to muscle group function and the joint segments the tendons cross.<sup>77</sup> A more clinically relevant assessment of both the extrinsic and intrinsic muscles may be accomplished by testing their function in foot shaping, stability, and force attenuation. For example, assessment of the patient's ability to maintain arches across loading conditions may yield more clinically relevant information on the synergy

of the posterior tibialis and the intrinsic muscles to maintain the medial longitudinal arch in both conditions. MMT of first metatarsal plantarflexion and adduction may yield more pertinent information on fibularis longus function as opposed to standard testing of foot eversion. Once deficits are identified, treatment that is specific to the impairment may be implemented. Treatments such as strengthening exercises, neuromuscular stimulation, biofeedback, and gait training may be employed with progressive loading for isolation and integration of the intrinsic and extrinsic muscles.<sup>78</sup> Video of some intrinsic foot exercises can be accessed at <https://goo.gl/ugffZ8>.

## CONCLUSIONS

In summary, the midfoot and forefoot are commonly injured and can be an insidious comorbidity in LAS and CAI. Overlooked physical impairment in the midfoot or forefoot may result in persistent limitation in function, disability, and/or impaired quality of life. It is clinically imperative for healthcare providers to assess and treat the ankle-foot complex as a whole, to include the midfoot and forefoot, even when symptoms are not manifest.

Examination and treatment of the midfoot and forefoot should complement the thorough examination and treatment of the proximal and distal tibiofibular, talocrural, and subtalar articulations typically performed following injury. Based on the prevalence, cost, morbidity and progression to CAI type symptoms develop at the same rate in isolated midfoot injury as it does in LAS,<sup>16</sup> the examination and treatment of the midfoot and forefoot may furnish additional pertinent information to the treating provider and allow for a more comprehensive plan of care. The midfoot may have a larger contribution to normal neurophysiologic and mechanical function than previously thought. Further research focused on investigating the role of multisegmented foot kinematics in individuals with LAS and CAI, development and validation of clinical tests of the midfoot and forefoot is suggested.

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